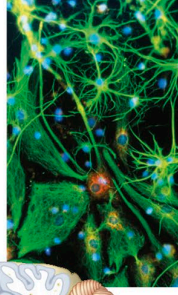
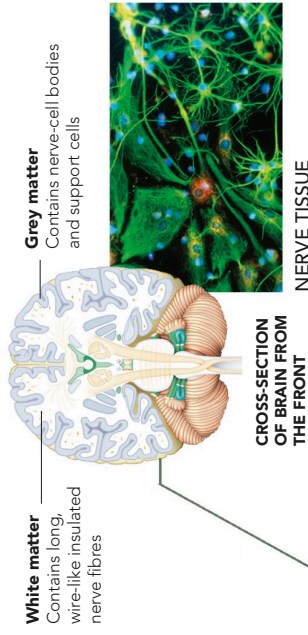


SPECIALIZED CELLS AND TISSUES

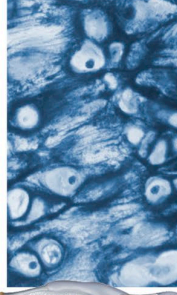
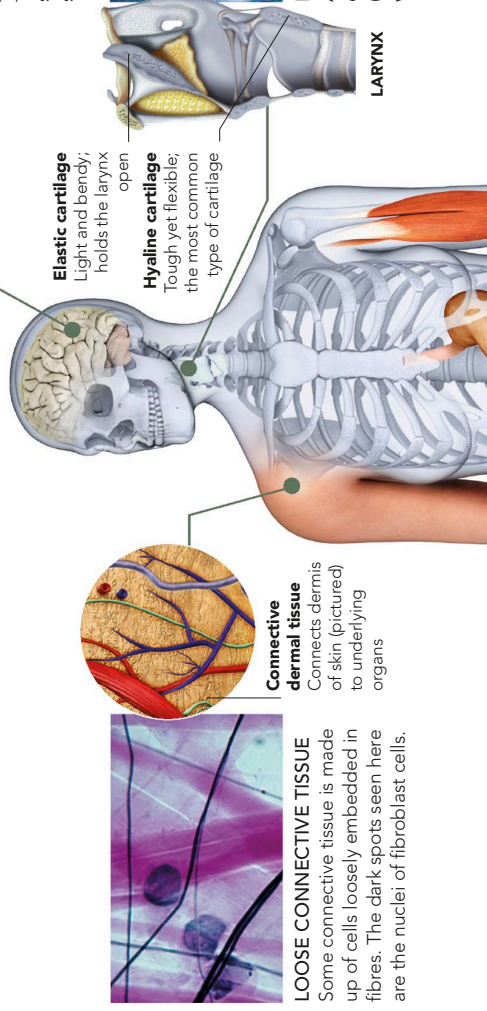
MORE THAN 200 TYPES OF SPECIALIZED CELLS POPULATE THE HUMAN BODY. EACH TYPE FORMS CLOSELY KNIT CONFIGURATIONS, WHICH ARE RECOGNIZABLE AS SPECIFIC TISSUES. IN SOME CASES, TISSUES ARE MADE OF SEVERAL TYPES OF CELLS.

TISSUE TYPES

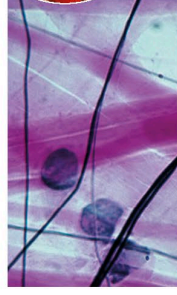
The cells that form tissue all have much the same structure and perform the same function. Classically there are four primary tissue types, derived from specific cell layers in the early embryo: epithelial, connective, muscle, and nerve. Blood, bone, cartilage, tendons, and ligaments are forms of connective tissue. The epidermis and the tissues that line almost every organ are all types of epithelial tissues. Muscle and nerve tissues, of course, form muscles and nerves.



A microscope image of nerve tissue showing the glial cells that support nerve cells (neurons). The spider-like forms supply nutrients.



A sample of cartilage from the epiglottis reveals round cells (chondrocytes) in fibres of elastin, which make it light and flexible.



Some connective tissue is made up of cells loosely embedded in fibres. The dark spots seen here are the nuclei of fibroblast cells.